



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2018 Year 11 Task 1

Thursday April 5, 2018

General instructions

- Working time – 70 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided.

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

- | | | |
|--|---|--------------------------------------|
| ☐ 11M3A – Mr Hwang | ☐ 11M3D – Dr Jomaa | ☐ 11M3G – Mr Lin |
| ☐ 11M3B – Ms Lee | ☐ 11M3E – Mr Berry | |
| ☐ 11M3C – Mr Ireland | ☐ 11M3F – Ms Ziazaris | |

Marker's use only.

QUESTION	1	2	3	4	5	6	7	Total	Total %
MARKS	$\overline{14}$	$\overline{7}$	$\overline{11}$	$\overline{10}$	$\overline{8}$	$\overline{9}$	$\overline{7}$	$\overline{66}$	$\overline{100}$

Question 1 (14 Marks)	Commence a NEW page.	Marks
(a) Write $0.1\dot{3}\dot{2}$ as a fraction in lowest terms, showing all working.		2
(b) Simplify $\sqrt[7]{x^2} \cdot \sqrt[3]{x}$.		2
(c) Factorise $6y^2 - 11y + 3$.		2
(d) Simplify giving your answer with positive indices		3
$\left(\frac{a^{-2}b}{a^3b^{-4}}\right)^{-2} \times \left(\frac{a^{-3}b^2}{a^{-1}b}\right)^3$		
(e) Simplify $\frac{x^3 - 1}{ax^2 - a} \div \frac{(x^2 + 1)^2 - x^2}{x^3 + 1}$.		3
(f) Rationalise $\frac{a\sqrt{b} + b\sqrt{a}}{\sqrt{a} + \sqrt{b}} - \frac{a\sqrt{b} - b\sqrt{a}}{\sqrt{a} - \sqrt{b}}$.		2

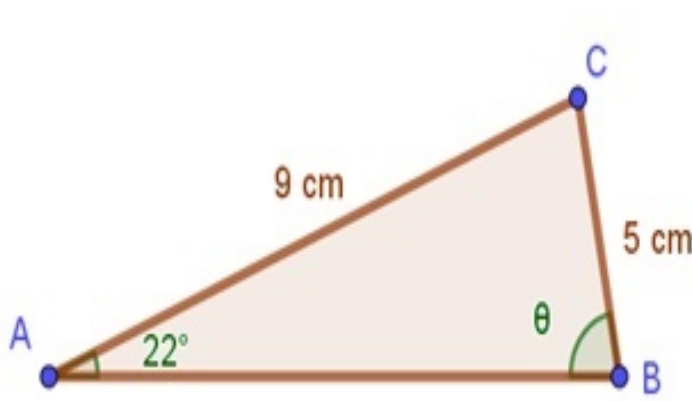
Question 2 (7 Marks)	Commence a NEW page.	Marks
Solve the following:		
(a) $ 2x - 1 = 7$		1
(b) $\frac{3}{x + 2} \leq x$		2
(c) $ 3x + 5 > 8$		2
(d) $ x + 4 = 2x - 5$		2

Question 3 (11 Marks)

Commence a NEW page.

Marks

- (a) Prove the identity: $\frac{1 + \sin t}{\cos t} = \frac{\cos t}{1 - \sin t}$ **1**
- (b) Solve $2 \tan(\theta - 60^\circ) = 0.9$, $0 \leq \theta \leq 360^\circ$ **2**
- (c) Find all possible values of θ in triangle ABC (see diagram). **2**



- (d) If $\cos \theta = \frac{1}{2}$ and $\tan \theta = -\sqrt{3}$, find $\sin \theta$ where $0 \leq \theta \leq 360^\circ$ **2**
- (e) i. Solve $3 \cos(3x) = 2$ where $0 \leq \theta \leq 360^\circ$ **2**
- ii. Hence, sketch $y = 2 - 3 \cos 3x$ for $0 \leq x \leq 360^\circ$, showing all important features. **2**

Question 4 (10 Marks)

Commence a NEW page.

Marks

- (a) Sketch graphs of the following curves indicating any intercepts and asymptotes.
- i. $y = \sqrt{3 - x}$ **2**
- ii. $y = 2^{2+x}$ **2**
- iii. $y = \frac{x + 2}{x^2 - 5x + 6}$ **2**
- (b) Given that
- $$f(x) = \begin{cases} x^2 - 2 & x < -1 \\ x^3 & -1 \leq x \leq 1 \\ -x^2 + 2 & x > 1 \end{cases}$$
- i. Find $f(-n) + f(0) + f(n)$, $n > 1$ **1**
- ii. Sketch the graph of $f(x)$, showing all the important features. **3**

Please Turn Over

- Question 5** (8 Marks) Commence a NEW page. **Marks**
- (a) Write down the domain and range of these functions:
- i. $y = x^2 + 8x + 11$ **2**
 - ii. $y = \frac{1}{\sqrt{16 - x^2}}$ **2**
- (b) Let $f(x) = 3x + 5$ and $h(x) = 3x^2 + 3x + 2$
- i. Find $f^{-1}(x)$ the inverse function of $f(x)$. **2**
 - ii. Find a function g such that $f(g(x)) = h(x)$ for all x . **2**

- Question 6** (9 Marks) Commence a NEW page. **Marks**
- (a) A is 3 km North and 4.5 km East of P . The Bearing of B from A is 155° . Find the distance PB , if the distance AB is 15 km. **4**
- (b) Consider a triangle ABC .
- i. Use the sine rule and the cosine rule simultaneously to prove that **3**

$$\cos^2 A + \sin^2 B + \sin^2 C = 1 + 2 \cos A \sin B \sin C$$
 - ii. Let $\angle A = 60^\circ$ and $\angle B = 45^\circ$. Use (i) to find the exact value of $\sin 75^\circ$. **2**

- Question 7** (7 Marks) Commence a NEW page. **Marks**
- (a) A right angled triangle ABC has an area of 25 cm^2 . P and h are the perimeter and the hypotenuse of ABC respectively.
- i. Show that $h = \frac{P^2 - 100}{2P}$ **2**
 - ii. Sketch the graph of the function h . **2**
- (b) Let $f(x)$ be a function defined as $f(0) = 0$ and $f(x) = f(x - 1) + 2x - 1$ for $x \geq 1$. Find an explicit formula for $f(x)$. **3**

End of paper.

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Solutions

Question 1

2 a) $0.1\dot{3}2$

Let $x = 0.1323232 \dots$

$$10x = 1.323232 \dots$$

$$1000x = 132.323232 \dots$$

$$990x = 131$$

$$x = \frac{131}{990}$$

2 b) $\sqrt[3]{x^2} \cdot \sqrt{x} = x^{2/3} \cdot x^{1/2}$
 $= x^{1\frac{1}{2}}$

2 c) $6y^2 - 11y + 3 = (3y - 1)(2y - 3)$

3 d) $\left(\frac{a^{-2}b}{a^3b^{-4}}\right)^{-2} \times \left(\frac{a^{-3}b^2}{a^{-1}b}\right)^3$
 $= \left(\frac{b^5}{a^5}\right)^{-2} \times \left(\frac{b}{a^2}\right)^3$
 $= \frac{a^{10}}{b^{10}} \times \frac{b^3}{a^6}$
 $= \frac{a^4}{b^7}$

3 e) $\frac{x^3 - 1}{ax^2 - a} \div \frac{(x^2 + 1)^2 - x^2}{x^3 + 1}$
 $= \frac{(x-1)(x^2+x+1)}{a(x+1)(x-1)} \div \frac{(x^2+1+x)(x^2+1-x)}{(x+1)(x^2-x+1)}$
 $= \frac{x^2+x+1}{a(x+1)} \times \frac{x+1}{x^2+x+1}$
 $= \frac{1}{a}$

$$2 f) \quad \frac{a\sqrt{b} + b\sqrt{a}}{\sqrt{a} + \sqrt{b}} - \frac{a\sqrt{b} - b\sqrt{a}}{\sqrt{a} - \sqrt{b}}$$

$$= \frac{(a\sqrt{b} + b\sqrt{a})(\sqrt{a} - \sqrt{b}) - (a\sqrt{b} - b\sqrt{a})(\sqrt{a} + \sqrt{b})}{a - b}$$

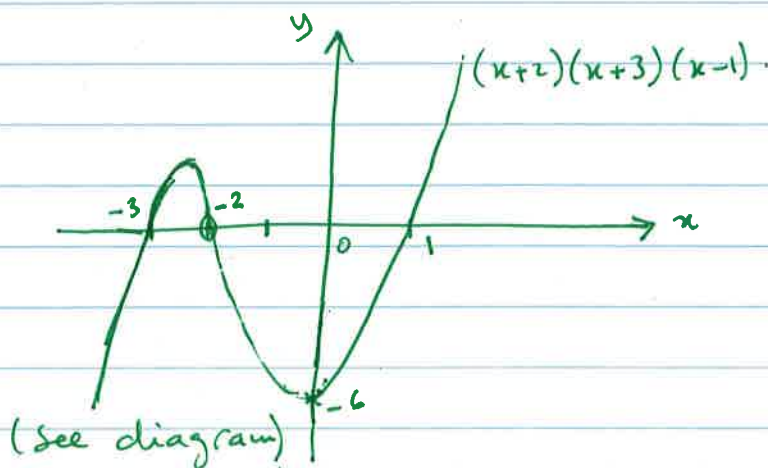
$$= \frac{a\sqrt{ab} + \cancel{ab} - \cancel{ab} - b\sqrt{ab} - (a\sqrt{ab} - \cancel{ab} + \cancel{ab} - b\sqrt{ab})}{a - b}$$

$$= 0$$

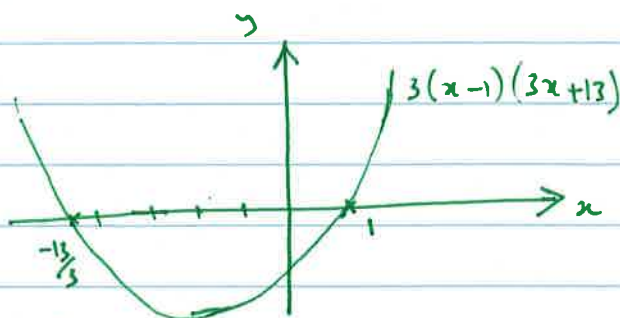
Question 2 :

(a) $|2x-1|=7$
 $\therefore (2x-1)^2 = 49$
 $\therefore (2x-1+7)(2x-1-7)=0$
 $\therefore (2x+6)(2x-8)=0$
 $\therefore x = -3 \text{ or } x = 4$

(b) $\frac{3}{x+2} \leq x$
 $\therefore 3(x+2) \leq x(x+2)^2$
 $\therefore (x+2)(3-x(x+2)) \leq 0$
 $\therefore (x+2)(-x^2-2x+3) \leq 0$
 $\therefore (x+2)(x^2+2x-3) \geq 0$
 $\therefore (x+2)(x+3)(x-1) \geq 0$
 $\therefore -3 \leq x < -2 \text{ or } x \geq 1$



(c) $|3x+5| > 8$
 $\therefore (3x+5)^2 > 64$
 $\therefore (3x+5+8)(3x+5-8) > 0$
 $\therefore (3x+13)(3x-3) > 0$
 $\therefore 3(x-1)(3x+13) > 0$
 $\therefore x < -\frac{13}{3} \text{ or } x > 1$

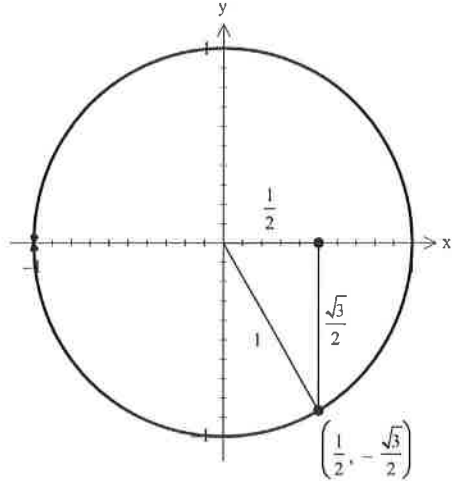


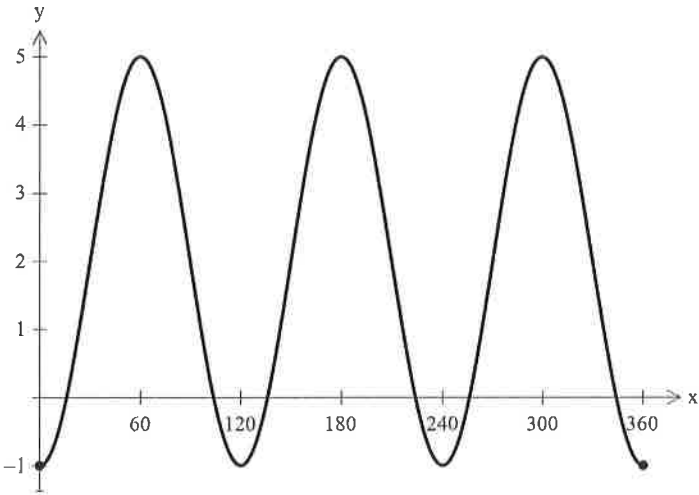
(d) $|x+4| = 2x-5$
 $\therefore (x+4)^2 = (2x-5)^2$
 $\therefore (x+4+2x-5)(x+4-2x+5) = 0$
 $\therefore (3x-1)(-x+9) = 0$
 $\therefore x = \frac{1}{3} \text{ or } x = 9$

check: $x = 9$, $|9+4| = 13$ and $2 \times 9 - 5 = 13$

$x = \frac{1}{3}$, $|\frac{1}{3}+4| = \frac{13}{3}$ and $2x-5 = \frac{2}{3}-5 = -\frac{13}{3}$ rejected

$\therefore x = 9$

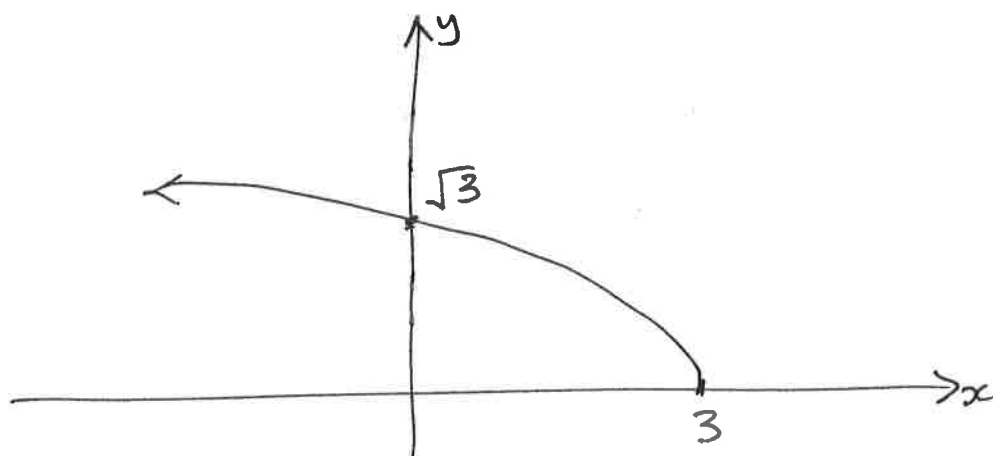
Q3	Solutions	Marks
(a)	$\text{LHS} = \frac{1 + \sin t}{\cos t} \times \frac{1 - \sin t}{1 - \sin t}$ $= \frac{1 - \sin^2 t}{\cos t (1 - \sin t)}$ $= \frac{\cos^2 t}{\cancel{\cos t} (1 - \sin t)}$ $= \frac{\cos t}{1 - \sin t}$ $= \text{RHS}$	✓ Correct proof clearly showing all steps
(b)	$0 \leq \theta \leq 360^\circ$ $-60^\circ \leq \theta - 60^\circ \leq 300^\circ$ $2 \tan(\theta - 60^\circ) = 0.9$ $\tan(\theta - 60^\circ) = 0.45$ $\theta - 60^\circ = 24^\circ 13' 40'' \text{ or } 204^\circ 13' 40''$ $\theta = 84^\circ 13' 40'' \text{ or } 264^\circ 13' 40''$ (to the nearest second)	✓ Acute answer ✓ Reflex answer
(c)	$\frac{\sin \theta}{9} = \frac{\sin(22^\circ)}{5}$ $\sin \theta = \frac{9 \sin(22^\circ)}{5}$ $\theta = 42^\circ 23' 57'' \text{ or } 137^\circ 36' 3''$ (to the nearest second)	✓ Acute answer ✓ Obtuse answer
(d)	 $\sin \theta = -\frac{\sqrt{3}}{2}$	✓ Correct value ✓ Correct quadrant

(e) (i)	$0 \leq x \leq 360^\circ$ $0 \leq 3x \leq 1080^\circ$ $3\cos(3x) = 2$ $\cos(3x) = \frac{2}{3}$ $3x = 48^\circ 11' 23'', 311^\circ 48' 37''$ $408^\circ 11' 23'', 671^\circ 48' 37''$ $768^\circ 11' 23'' \text{ or } 1031^\circ 48' 37''$ $x = 16^\circ 3' 48'', 103^\circ 56' 13''$ $136^\circ 3' 48'', 223^\circ 56' 13''$ $256^\circ 3' 48'' \text{ or } 343^\circ 56' 13''$ (to the nearest second)	✓ Any one correct answer ✓ All correct answers
(e) (ii)		✓ Amplitude and period ✓ Vertical translation with the x intercepts in reasonable positions

Q4

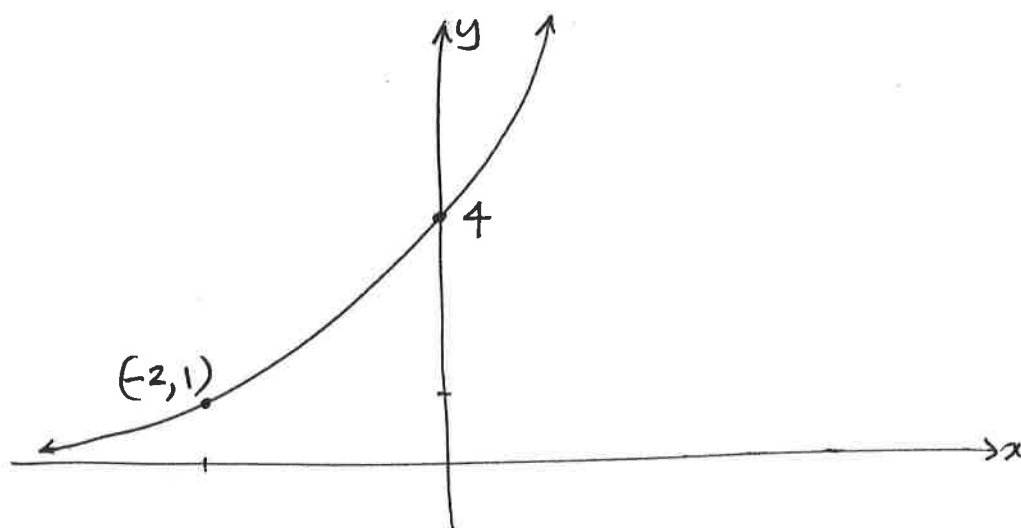
(a)

(i) $y = \sqrt{3-x}$



✓✓

(ii) $y = 2^{2+x} = 4 \times 2^x$



✓✓

Q4 (a) (ctd)

$$(iii) \quad y = \frac{x+2}{x^2-5x+6} = \frac{x+2}{(x-2)(x-3)}$$

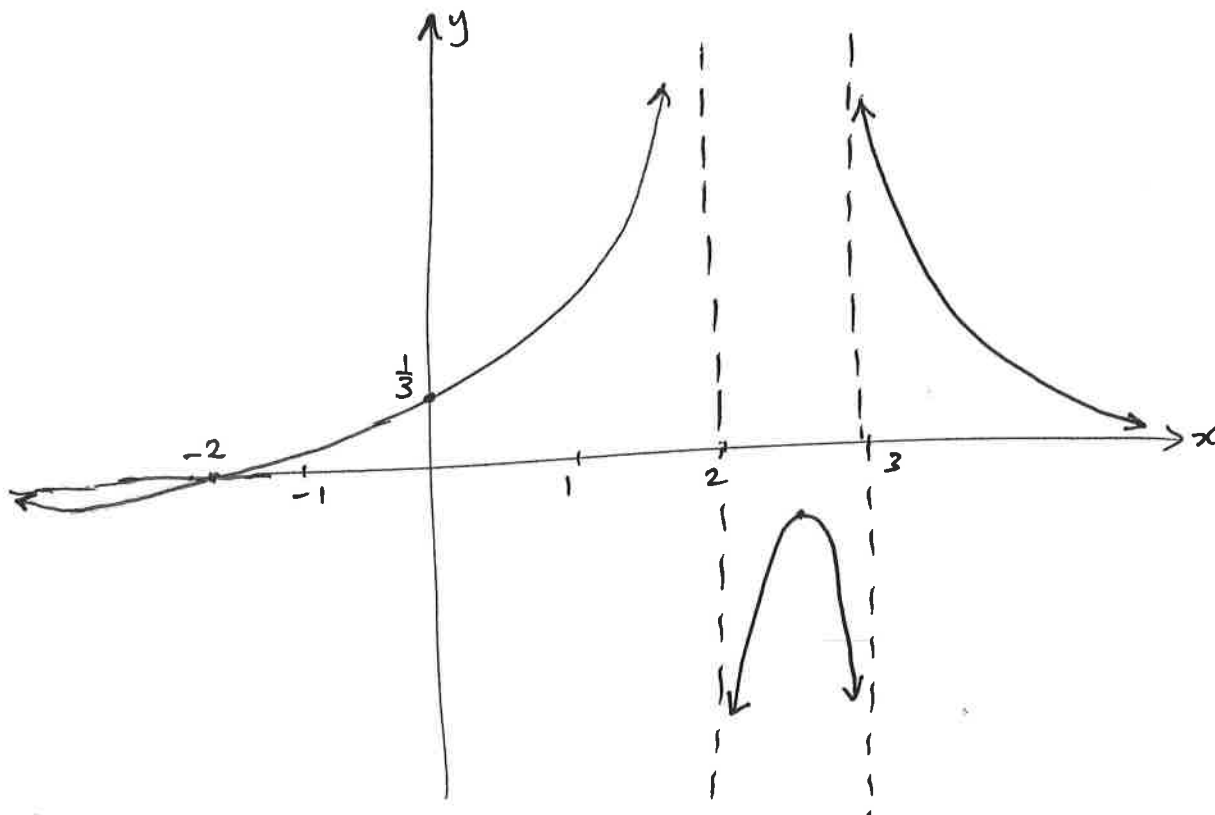
vert. asymptotes: $x=2, x=3$

y-int. = $\frac{1}{3}$, x-int. = -2

as $x \rightarrow \infty, y \rightarrow 0^+$. as $x \rightarrow -\infty, y \rightarrow 0^-$.

x	-3	-2	0	2	$2\frac{1}{2}$	3	4
y	-	0	+	*	-	*	+

↑ Table of signs.



[Note: local max
between 2 & 3 is at
 $x = 2.5 - 2 \div 2.47$
not at 2.5]

Full marks needs all
asymptotes (labelled)
and y-intercept.

✓✓

Q4 (b)

$$f(x) = \begin{cases} x^2 - 2, & x < -1 \\ x^3, & -1 \leq x \leq 1 \\ -x^2 + 2, & x > 1 \end{cases}$$

$$(i) \quad f(-n) = (-n)^2 - 2 = n^2 - 2 \quad (\text{as } n > 1)$$

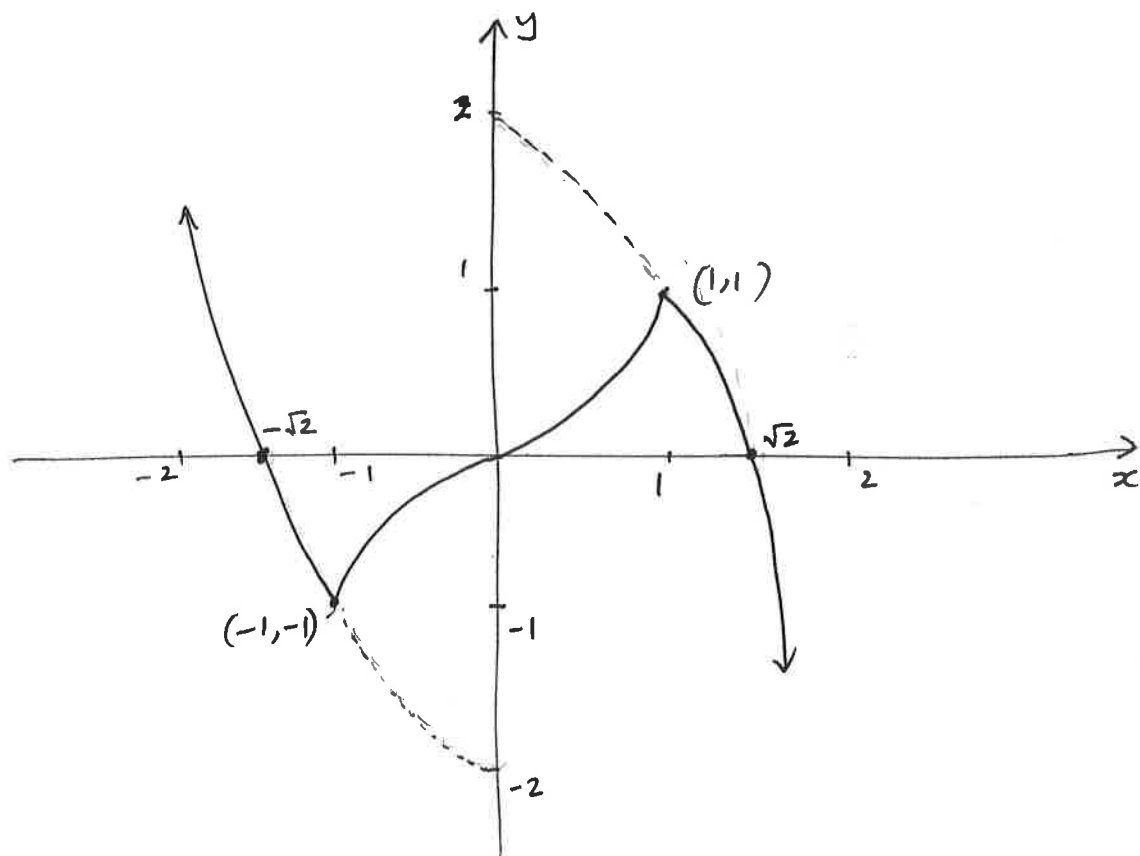
$$f(0) = 0^3 = 0$$

$$f(n) = -n^2 + 2$$

$$\therefore f(-n) + f(0) + f(n) = n^2 - 2 + 0 - n^2 + 2 = 0$$

✓

(ii)



✓✓✓

Question 5

a) i) $y = x^2 + 8x + 11$

vertex at $(-4, -5)$ and concave up

$\therefore D: x \text{ is all real}$

$R: y \geq -5$

ii) $D: -4 < x < 4$

$R: y \geq \frac{1}{4}$

b) i) Inverse relation :

$$x = 3y + 5$$

$$\therefore 3y = x - 5$$

$$y = \frac{x-5}{3}$$

$$\therefore f^{-1}(x) = \frac{x-5}{3}$$

ii) $f(g(x)) = 3x^2 + 3x + 2$

$$3g(x) + 5 = 3x^2 + 3x + 2$$

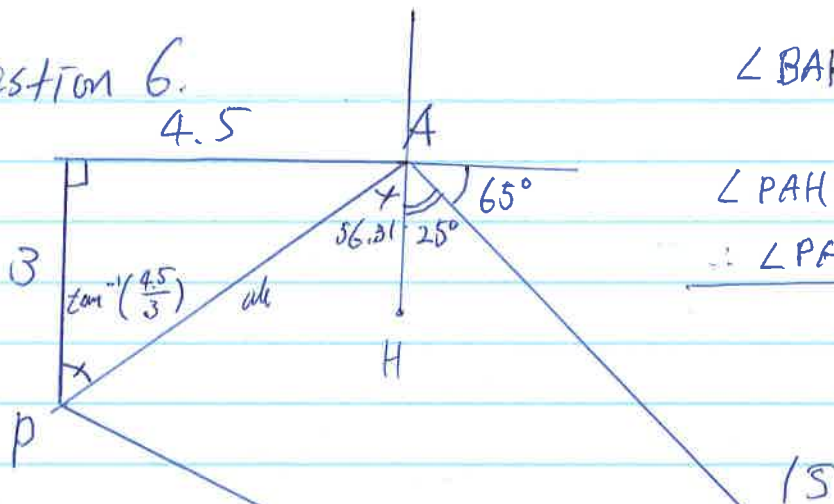
$$3g(x) = 3x^2 + 3x - 3$$

$$g(x) = x^2 + x - 1$$

Question 6.

4.

a)



$\angle BAH = 25^\circ$ (from bearing of B from A)

$$\angle PAH = \tan^{-1}\left(\frac{4.5}{3}\right) \approx 56.31^\circ$$

$$\therefore \angle PAB = 81.31^\circ$$

$$PA = \sqrt{4.5^2 + 3^2} = \sqrt{29.25} \approx 5.408 \text{ (3 d.p.)}$$

$$PB = \sqrt{15^2 + 29.25 - 2 \times 15 \times \sqrt{29.25} \times \cos(81.31^\circ)} = 15.16 \text{ km (2 d.p.)}$$

b) i) $\cos^2 A + \sin^2 B + \sin^2 C = 1 + 2 \cos A \sin B \sin C$

from sine rule. $a = b \times \frac{\sin A}{\sin B}$ $b = b$ $c = b \times \frac{\sin C}{\sin B}$

sub into cos rule.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\frac{b^2 \times \sin^2 A}{\sin^2 B} = \frac{b^2 + b^2 \times \frac{\sin^2 C}{\sin^2 B} - 2b^2 \frac{\sin C}{\sin B} \cos A}{\sin^2 B} \quad \left| \div b^2 \text{ \& } \times \cos^2 B \right.$$

$$\sin^2 A = \sin^2 B + \sin^2 C - 2 \sin B \sin C \cos A \quad \left| \sin B \sin C \right.$$

$$= 1 - \cos^2 A$$

$$1 - \cos^2 A = \sin^2 B + \sin^2 C - 2 \cos A \sin B \sin C$$

$$1 + 2 \cos A \sin B \sin C = \cos^2 A + \sin^2 B + \sin^2 C$$

as given.

~~1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11~~
~~12, 13, 14, 15~~



Q6.

(2) b) ii) $\cos^2 60^\circ + \sin^2 45^\circ + \sin^2 75^\circ = 1 + 2 \cos 60^\circ \sin 45^\circ \sin 75^\circ$
 let $\sin 75^\circ = x$.

$$\frac{1}{4} + \frac{1}{2} + x^2 = 1 + 2 \times \frac{1}{2} \times \frac{\sqrt{2}}{2} \times x$$

$$x^2 - \frac{\sqrt{2}}{2}x - \frac{1}{4} = 0 \quad | \times 4$$

$$4x^2 - 2\sqrt{2}x - 1 = 0$$

Using
quadratic
formula

$$x = \frac{2\sqrt{2} \pm \sqrt{8+16}}{8} = \frac{2\sqrt{2} \pm 2\sqrt{6}}{8}$$

$$= \frac{\sqrt{2} \pm \sqrt{6}}{4}$$

$$\therefore \sin 75^\circ = \frac{\sqrt{2} + \sqrt{6}}{4}$$

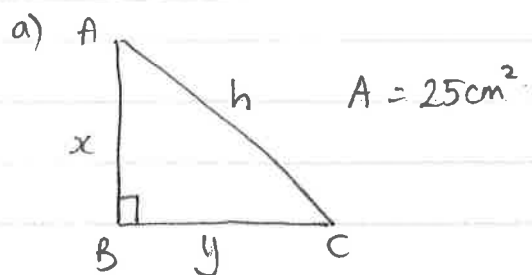
$$\text{or } \frac{\sqrt{2} - \sqrt{6}}{4}$$

but $\sin 75^\circ$ is acute $\sin 75^\circ$ has to be positive

$$\therefore \sin 75^\circ = \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$= 0.9659$$

Question 7 Yr 11 2018 Task 1. Ext 1



(i) $P = x + y + h$

$A = \frac{1}{2}xy$

$x^2 + y^2 = h^2 \quad \text{--- (2)}$

$25 = \frac{1}{2}xy$

$xy = 50 \quad \text{--- (1)}$

$\therefore P - h = x + y$

$(P - h)^2 = (x + y)^2$

$(P - h)^2 = x^2 + 2xy + y^2$

Using --- (1) & (2)

$(P - h)^2 = h^2 + 100$

$P^2 - 2Ph + h^2 = h^2 + 100$

$P^2 - 100 = 2Ph$

$h = \frac{P^2 - 100}{2P}$

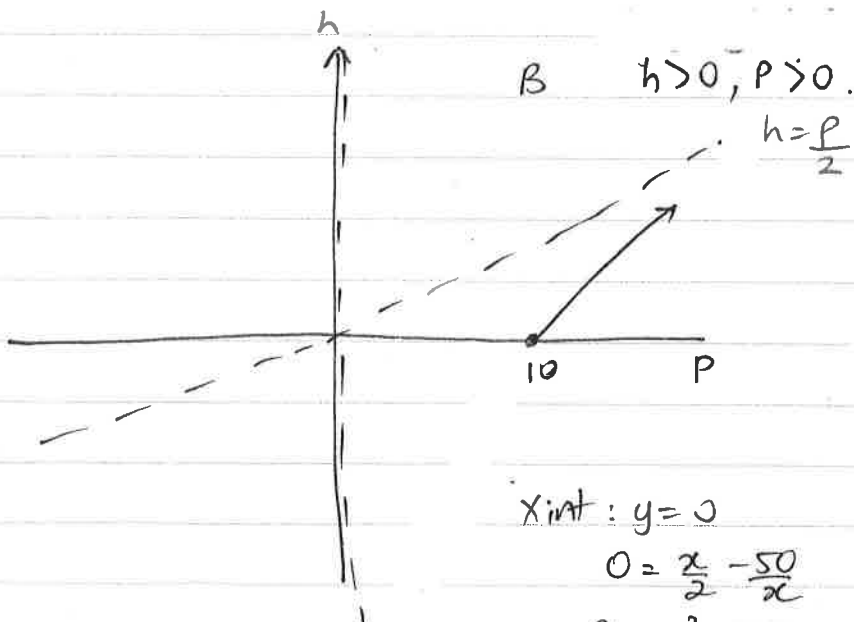
(ii) $h = \frac{P^2 - 100}{2P}$

$h = \frac{P^2}{2P} - \frac{100}{2P}$

$h = \frac{P}{2} - \frac{50}{P}$

i.e. $y = \frac{x}{2} - \frac{50}{x}$

Asymptotes $y = \frac{x}{2}$
 $x = 0$



Xint: $y = 0$

$0 = \frac{x}{2} - \frac{50}{x}$

$0 = x^2 - 100$

$x = 10$

$$b) f(0) = 0$$

$$f(1) = f(0) + 1$$

$$= 0 + 1$$

$$= 1$$

$$f(2) = f(1) + 3$$

$$= 1 + 3$$

$$= 4$$

$$f(3) = f(2) + 5$$

$$= 4 + 5$$

$$= 9$$

x	0	1	2	3
$f(x)$	0	1	4	9

$$f(4) = f(3) + 7$$

$$= 9 + 7$$

$$= 16$$

$$\therefore \boxed{f(x) = x^2}$$

OR

$$f(x) = f(x-1) + 2x-1$$

$$(x-1)^2 = x^2 - 2x + 1$$

$$\therefore f(x) = (x-1)^2 + 2x-1$$

$$= x^2 - 2x + 1 + 2x - 1$$

$$= x^2$$

Check

$$f(0) = 0$$

f